

RACHEL RATHBONE

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EXPERIENCED ANALYTICAL CHEMIST

PROFESSIONAL PROFILE

- Self motivated chemist with a solid background in organic chemistry, spectroscopy, instrumentation and research and development. Thrives in a challenging environment while maintaining creative ingenuity.
- Experienced technician in the fields of electronics and telecommunications. Fluent in calibration, maintenance, monitoring, repair, testing, troubleshooting and quality assurance of a broad range of equipment.
- Level three CRLA certified tutor with three years of experience as a Supplemental Instructor for Chemistry and Biology at Lone Star University.

AREAS OF EXPERTISE

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|---------------------------------------|-----------------------------|--------------------------------|
| • Infrared Spectroscopy | • Flow Cytometry | • Polymerase Chain Reactions |
| • Nuclear Magnetic Resonance | • Cyclic Voltammetry | • Quality Assurance Inspection |
| • AFM, SEM and STM | • Organic Techniques | • Maintenance Supervision |
| • Mass Spectrometry | • Gram Staining | • Technical Manuals /Writing |
| • Confocal Microscopy | • DNA/RNA Extraction | • Schematic Reading |
| • Spectrophotometry | • Gel Electrophoresis | • Troubleshooting and Repair |
| • Chromatography | • Oligonucleotide Synthesis | • Job Planning |
| • Spectroscopy (multiple disciplines) | • Western Blot | • Training of Personnel |
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PROFESSIONAL EXPERIENCE

Supplemental Instructor - Chemistry and Biology, 08/2010 to 05/2013

Lone Star College - Kingwood, TX

- Organized and supervised peer-assisted study sessions for academically challenging courses in which students developed study skills, compared and refined notes, discussed readings and developed organizational tools.
- Served as a liaison between professor and student while regularly performing SI duties.
- Coordinated with the professor to create lesson plans and demonstrations specifically designed for the task at hand.
- Gathered and evaluated data on students and professor performance to increase overall academic success.
- Performed inspections, testing, maintenance repair, and calibration of various chemical instruments.
- Instructed students on the theory and proper operation of analytical instrumentation.
- Enforced laboratory safety practices in accordance with Standard Operating Procedures.
- Obtained a College Reading and Learning Association Level III Tutor Certification.

Research Assistant, 06/2010-09/2010 and 06/2011- 09/2011

Rice University - Houston, TX

- Performed experiments and analyzed samples to provide information on compounds in question.
- Performed qualitative and/or quantitative analyses of data, using computer software; such as Matlab and LabVIEW as applicable.
- Provided assistance with the preparation of project-related reports, manuscripts, and presentations.
- Gained invaluable experience in chromatography, flow cytometry, and techniques; FCS, IR, NMR, ICP and GCMS.

- Designed and built a laser system that combined both widefield and TIRF microscopy.
- Conducted extensive studies on polymers and PEGylated surface interactions.

Network Operations Command Center Technician, 09/2006 to 10/2007

Alltel Wireless - Mankato, MN

- Enabled proactive monitoring, maintenance, testing and troubleshooting of servers in a nationwide network.
- Coordinated with technicians and local authorities on CDMA, TDMA, Clearwave equipment and servers.
- Opened trouble tickets when a customer issue/notification/request was received or network outage or degradation was observed.
- Coordinated troubleshooting efforts with the NOCC manager, assistant manager and/or support engineers to isolate network issues.

Aircraft Electrical/Instrument/Flight Control Systems Laboratory Technician, 05/2001-05/2006

United States Marine Corps - San Diego, CA

- Performed inspections, testing, maintenance repair, and calibration of components, assemblies, sub-assemblies, modules, cards, printed circuit boards and ancillary equipment.
- Responsible for maintaining all changes to technical publications containing testing procedures and schematics for all systems.
- Responsible for inventory, maintenance and calibration schedules of millions of dollars in test equipment amounting to over 200 items.
- Served as the day crew supervisor and was responsible for oversight and direction of the workload as well as a crew of 30 personnel daily.
- Served as a collateral duty inspector and was responsible for all events from the start of the maintenance operation to its completion.
- Served as a quality assurance inspector who monitored safety, conducted surveillance/spot checks and work center audits as directed by the QA Manager. Possessed in-depth knowledge of Quality Assurance procedures, technical manuals and directives, safety procedures, support equipment and special tools.
- Researched and interpreted drawings and papers as well as wrote reports with clarity and technical accuracy.
- Served as a liaison between the laboratory and other technical representatives pertaining to issues with all equipment, test sets and publications.

EDUCATION

Bachelor of Science in Chemistry, Minor in Biology - University of Houston- Houston, TX 12/2014

Associate of Science in Chemistry - Lone Star University- Kingwood, TX 05/2012

Rice Quantum Institute- Undergraduate Research - Rice University 06/2011- 09/2011

Research Experience for Undergraduates- Rice University 06/2010-09/2010

Certified Fiber Optics Installer - Aviation Electronic Schools of America- 05/2005

Rotary Wing Electrical Systems Intermediate Maintenance Certification - Air Maintenance Training Detachment- Camp Pendleton, CA 06/2002

Aviation Electrician's Mate Certification- Air Technical Training Center - Pensacola, FL 01/2002

ACCOMPLISHMENTS

Third Author - Publication of Research: Charlisa R. Daniels, R. K. (2011). Permeability of anti-fouling PEGylated surfaces probed by fluorescence correlation spectroscopy. *Colloids and Surfaces B: Biointerfaces*, 31-38.

NSF Engineering Education Award- 2011

CRC Freshman Chemistry Award - 2010